

WFH Burnout Predictor

Predicting work-from-home burnout score from daily behavioral signals.

A CRISP-DM regression and classification project, evaluated on employees unseen during training.

Can daily behavior predict burnout before it peaks?

Predicting `burnout_score` from work-from-home indicators such as workload, screen time, meetings, breaks, after-hours work, sleep and task completion; evaluated on employees not seen during training.

Key design decisions

- Grouped train / test split: no employee appears on both sides.
- Baselines first: dummy models set the bar before complex ones.
- Explicit limits: claims are scoped to what the data supports.



Main task

Regression on `burnout_score`: the continuous burnout level for each employee.



Secondary task

Classification of `burnout_risk` (Low / Medium / High), used mainly for interpretation.



Focus

The dataset is technically simple, so the project emphasizes honest data and result understanding.



Business
Understanding



Data
Understanding



Data
Preparation



Modeling



Evaluation



Deployment

A clean, repeated-measures dataset

Each of the 180 employees contributes 10 daily records, so the identifier is useful for evaluation design (grouping train and test) but not as a predictive feature.



1,800

observations



180

employees (users)



10

records per user



0

missing values

Predictors — 8 daily behavioral signals

day_type

work_hours

screen_time_hours

meetings_count

breaks_taken

after_hours_work

sleep_hours

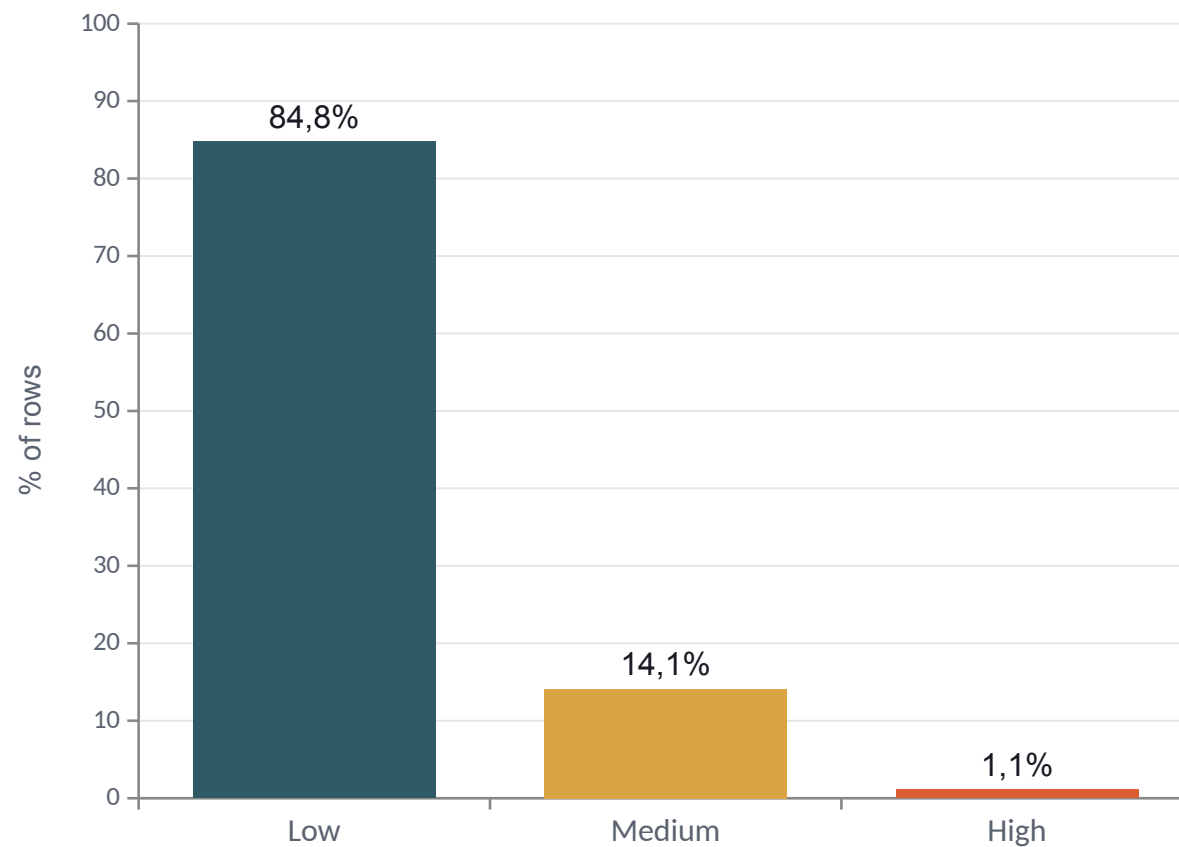
task_completion_rate

Targets

- **burnout_score** continuous · regression target
- **burnout_risk** Low / Medium / High · classification target

Technical quality is high: no missing values, no duplicates, valid ranges. But external representativeness beyond this public sample remains unverified.

burnout_risk leans heavily toward Low



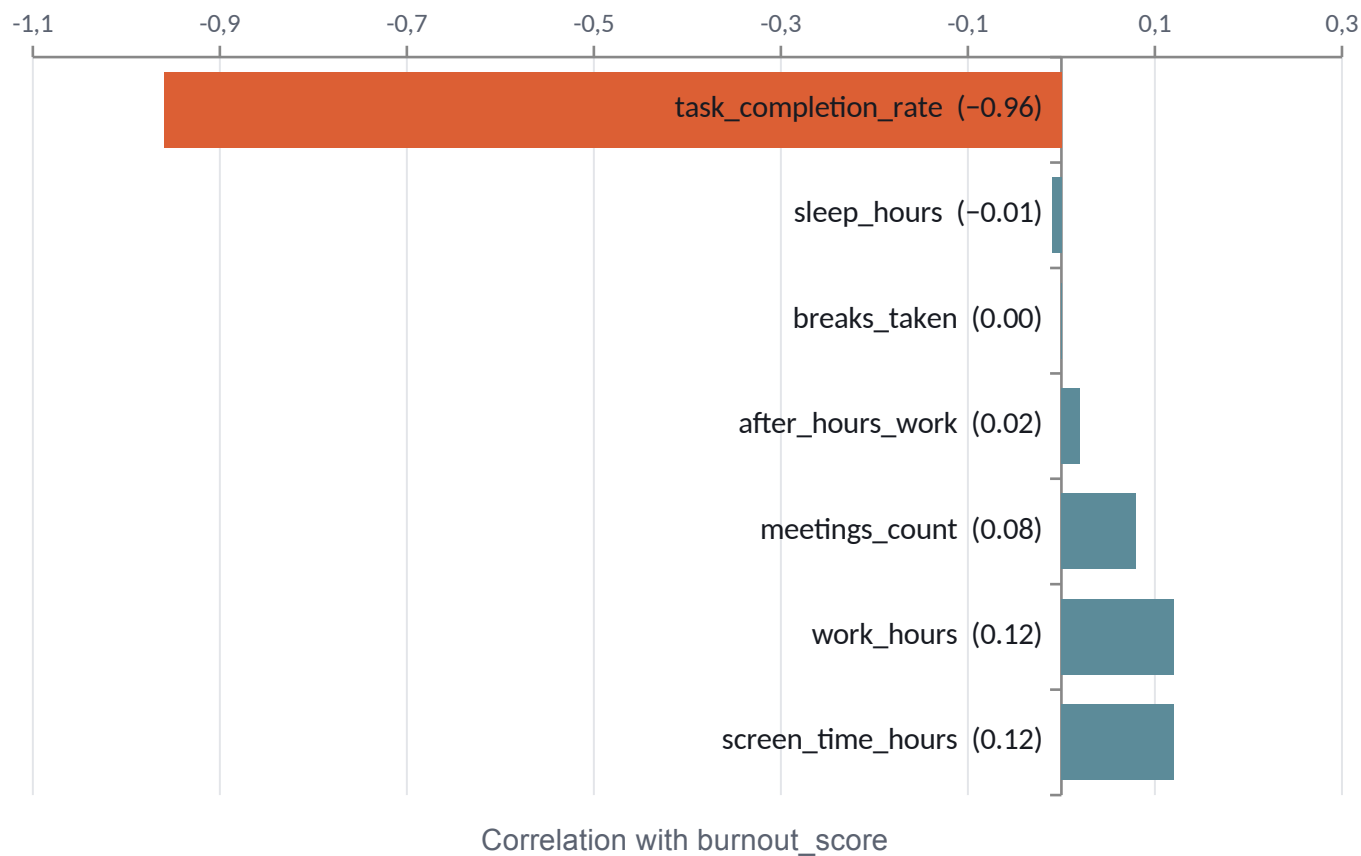
Why this matters

- Low-risk rows dominate the dataset — 1,527 of 1,800 rows.
- High risk appears in only 20 records — about 1.1%.
- A classifier can look accurate while missing the rarest class entirely.
- burnout_score is treated as the main target: it drives the regression task and, through fixed thresholds, the risk class.
burnout_risk is a deterministic bucketing: Low ≤ 69.92 | Medium 70.01–109.99 | High ≥ 110.22 — non-overlapping bands, verified programmatically.

20 rows

out of 1,800 carry the High-risk label

One feature carries almost all the signal



-0.96

Correlation between task_completion_rate and burnout_score.

This is useful for prediction, but it is also the main warning sign of the project: the feature may sit very close to the logic that generated the target in the first place.

A univariate Ridge on task_completion_rate alone reaches test $R^2=0.934$; the full 13-feature model achieves $R^2=0.936$.

All other predictors stay within ± 0.12 — a much weaker, more plausible behavioral signal.

The split is designed around people, not rows

If the same employee appeared in both train and test, the model could memorize personal baselines instead of learning generalizable patterns.



Excluded

`user_id` is never used as a predictor — it has no generalizable meaning.

1,440 / 360

train / test rows



Grouped split

Records from the same employee stay entirely in train or entirely in test.

144 / 36

train / test users



Pipeline

Encoding, scaling and model fitting all run inside scikit-learn pipelines.

0

overlapping users

Engineered features: interpretable workload-pressure signals

work_intensity

$\text{work_hours} \div \text{breaks_taken} \rightarrow \text{net work pressure}$

meeting_density

$\text{meetings} \div \text{work_hours} \rightarrow \text{interruptions per hour}$

screen_work_ratio

$\text{screen_time} \div \text{work_hours} \rightarrow \text{display exposure rate}$

sleep_deficit

$\max(0, 7 - \text{sleep_hours}) \rightarrow \text{hours below 7h threshold}$

overwork_flag

1 if `work_hours` > 8, else 0

Baselines first, then increasingly complex models



Baseline

DummyRegressor predicts the training-set median.
Sets the performance floor: CV MAE 18.49. Any model failing to beat this is useless.



Linear

Ridge adds L2 regularisation to prevent unstable coefficients when features correlate.
The CV MAE 5.06 confirms the signal is mostly linear.



Non-linear

- Random Forest: 300 trees on bootstrap samples, averaged for lower variance.
- Hist Gradient Boosting: sequential trees, each correcting the previous residuals.



Validation

5-fold GroupKFold keeps user grouping intact inside each fold using same constraint as the main split, preventing any employee from leaking across validation boundaries.

Cross-validation comparison (training set, GroupKFold)

Model	CV MAE	CV R ²
Random Forest	4.89	0.925
Ridge	5.06	0.916
Hist Gradient Boosting	5.13	0.918
KNN	9.05	0.742
Dummy (median)	18.49	-0.049

Random Forest wins on CV (MAE 4.89) and is selected for tuning.

KNN scores 9.05: with 13 scaled features, training points become nearly equidistant and local neighbourhoods lose predictive meaning.

All meaningful models clear the dummy baseline

Model	MAE	RMSE	R ²
Ridge	4.79	6.31	0.934
Random Forest	4.85	6.41	0.932
Hist Gradient Boosting	5.14	6.73	0.925
KNN	9.26	11.65	0.776
Dummy (median)	20.35	25.80	-0.099

Test-set results before hyperparameter tuning.

TUNED MODEL

Random Forest Regressor

4.84

Test MAE

6.26

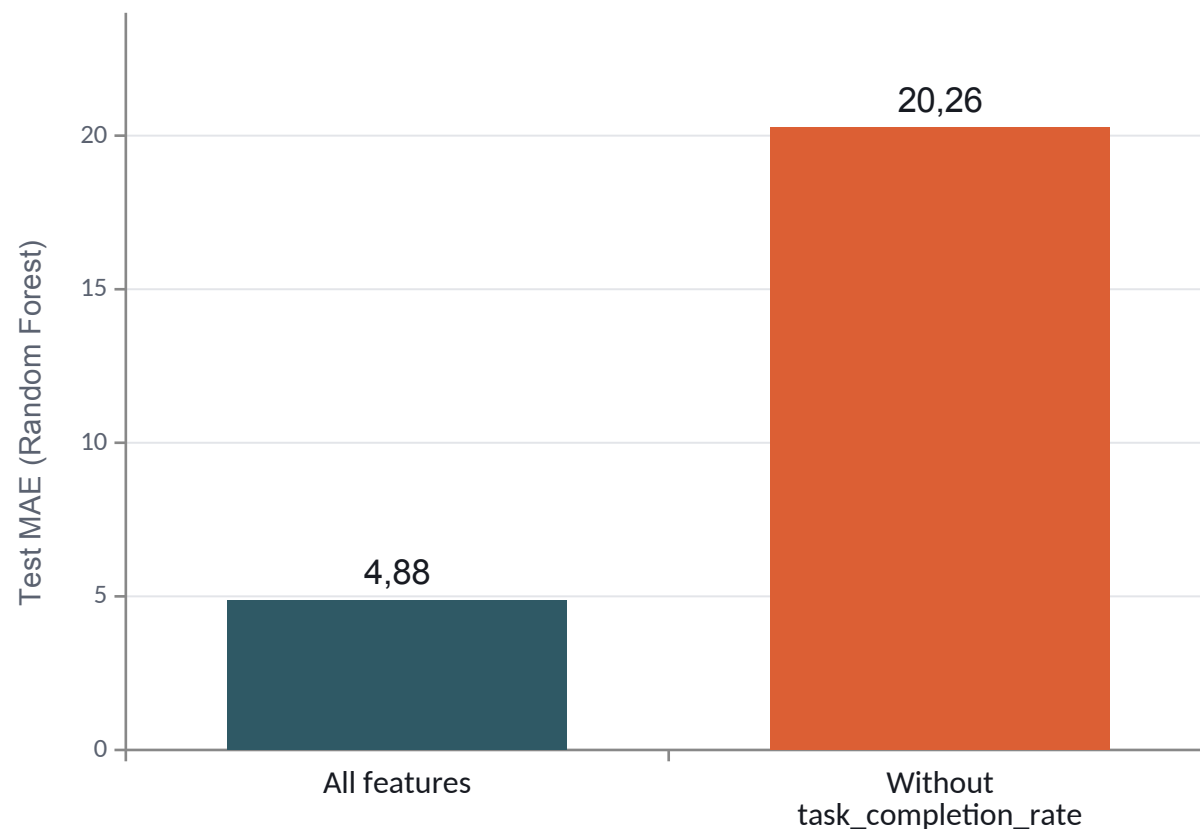
Test RMSE

0.935

Test R²

Params: n_estimators=200 · max_features=0.7 ·
min_samples_leaf=4 · max_depth=None.

Remove one feature, and performance collapses



task_completion_rate alone carries almost all of the predictive power.

Dropping it pushes Random Forest MAE from 4.85 to 20.26, within noise of the dummy median baseline (≈ 20.35).

This does not invalidate the pipeline.

It narrows the claim: the project demonstrates predictive modeling and is explicit about exactly where its reliability comes from.

Risk labels help, but the rarest class stays hard to catch

0.700

Balanced accuracy

0.690

Macro F1

**Random Forest
balanced**

Best classifier

Risk class	Precision	Recall	F1	Support
Low	0.99	0.95	0.97	294
Medium	0.75	0.90	0.82	62
High	0.33	0.25	0.29	4

Classifier selected by macro F1, which weights all three classes equally.
Random Forest balanced wins (macro F1 0.690 vs 0.646 for Logistic Regression).

Note: LR had higher High-risk recall (0.75 vs 0.25): it caught 3 of 4 true High cases while RF catches only 1.

This is a genuine trade-off: for a screening tool, missing a real burnout case carries higher cost than a false alarm, making High recall operationally critical alongside macro F1.
All High-class figures rest on 4 test cases and should be read as directional only.

A cautious decision-support tool, not an automatic verdict



Interpretability

Permutation importance confirms
task_completion_rate dominates the prediction,
but it should not be read causally.



Fairness checks

The dataset has no classic sensitive attributes.
Errors and high-risk recall are still checked across
day_type, after_hours_work and work-hour
bands.

Note: High-risk recall per subgroup is computed on 0–3 positive cases so figures indicate direction, not statistical certainty.



Deployment plan

Save the full pipeline, not just the model.
Monitor feature and performance drift.
Repeat leakage and fairness audits over time.



External validation on real organizational data is required before any operational use — this remains a research-stage pipeline.

What this project actually demonstrates

01

Data understanding drives the project

Clean but simple data, repeated users, and a rare High-risk class shape every later choice.

02

Evaluation is conservative

Users are grouped end-to-end, so no employee leaks between training and test.

03

The regression result is strong

Tuned Random Forest reaches MAE 4.84 and R^2 0.935 on employees unseen during training.

04

The main limitation is explicit

Performance leans heavily on `task_completion_rate`.

05

Final read

A transparent predictive pipeline but there is not evidence of causal burnout mechanisms.